

Database Systems

Unit – 1 Introduction

Instructor
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Text & Reference Books

Text Book

- Database Systems – Design, Implementation and Management: Carlos Coronel, Steven Morris (12th Ed.)

Reference Books

A.Modern Database Management: Jeffery. A. Hoffer (10th Ed.)

B.Fundamentals of Database Systems: Elmasri (7th Ed.)

Course Information

Assessment Category	Marks
Quizzes	8
Assignments	7
Project + Presentations	10

Course Outline

- Introduction
- Database Architecture and Components
- Database Development and Data Modeling
- Relational Data Model
- Structured Query Language (SQL)
- Entity Relationship (E-R) Model
- Normalization
- Structured Query Language (SQL)
- Database Design
- Transaction Management and Concurrency Control
- Advanced Topics and Presentations

Lecture Outline

- Data? Information? Database?
- File Processing System (FPS)
- Drawbacks of FPS
- Database System or Integrated Database Environment(IDE)
- Advantages of IDE
- Types of Databases

Data

- Data refers to the raw facts & figures concerning:
 - ✓ PEOPLE
 - ✓ EVENT
 - ✓ ORGANISATION
 - ✓ OBJECT
- The amount of available data in the modern computer world is literally exploding.

Data

- Large volume of facts, difficult to interpret or make decisions based on

Class Roster

Course: MGT 500
Business Policy

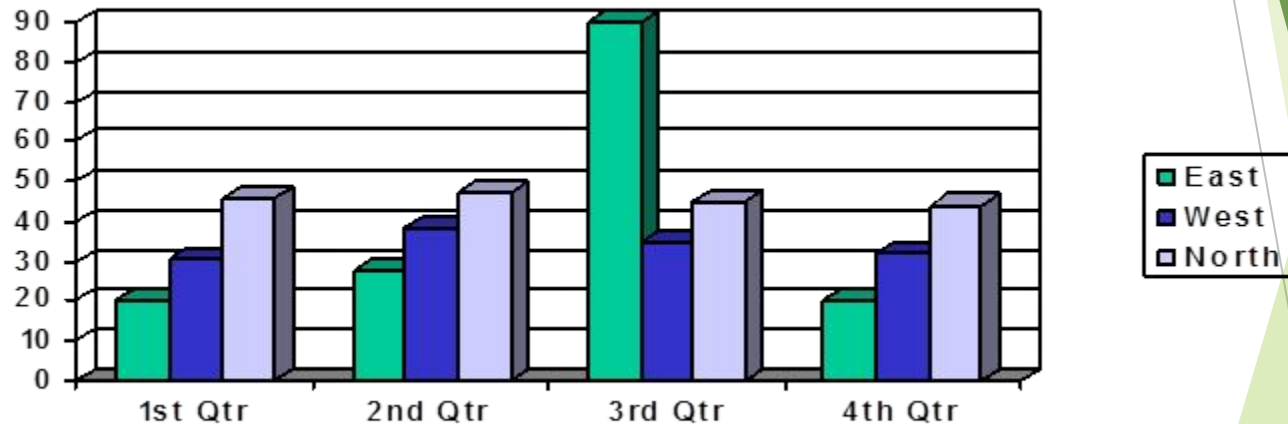
Semester: Spring 200X

Section: 2

<u>Name</u>	<u>ID</u>	<u>Major</u>	<u>GPA</u>
Baker, Kenneth D.	324917628	MGT	2.9
Doyle, Joan E.	476193248	MKT	3.4
Finkle, Clive R.	548429344	PRM	2.8
Lewis, John C.	551742186	MGT	3.7
McFerran, Debra R.	409723145	IS	2.9
Sisneros, Michael	392416582	ACCT	3.3

Data vs Information

- Information is derived from data OR Information is the processed data
- Information may be presented either Textually or Graphically or both



- The representation and amount of information depends upon the requirements of a user.

Record Keeping Techniques

1. Manual Record Keeping.
2. Computerized Record Keeping
 - a) File Processing System.
 - b) Database System or Integrated Database Environment. (The paradigm shift)

Manual System - Drawbacks

- High data volume
- Not reliable
- Inefficient
- Duplication of data
- Inconsistency
- The System can't answer complex queries involving multiple departments.

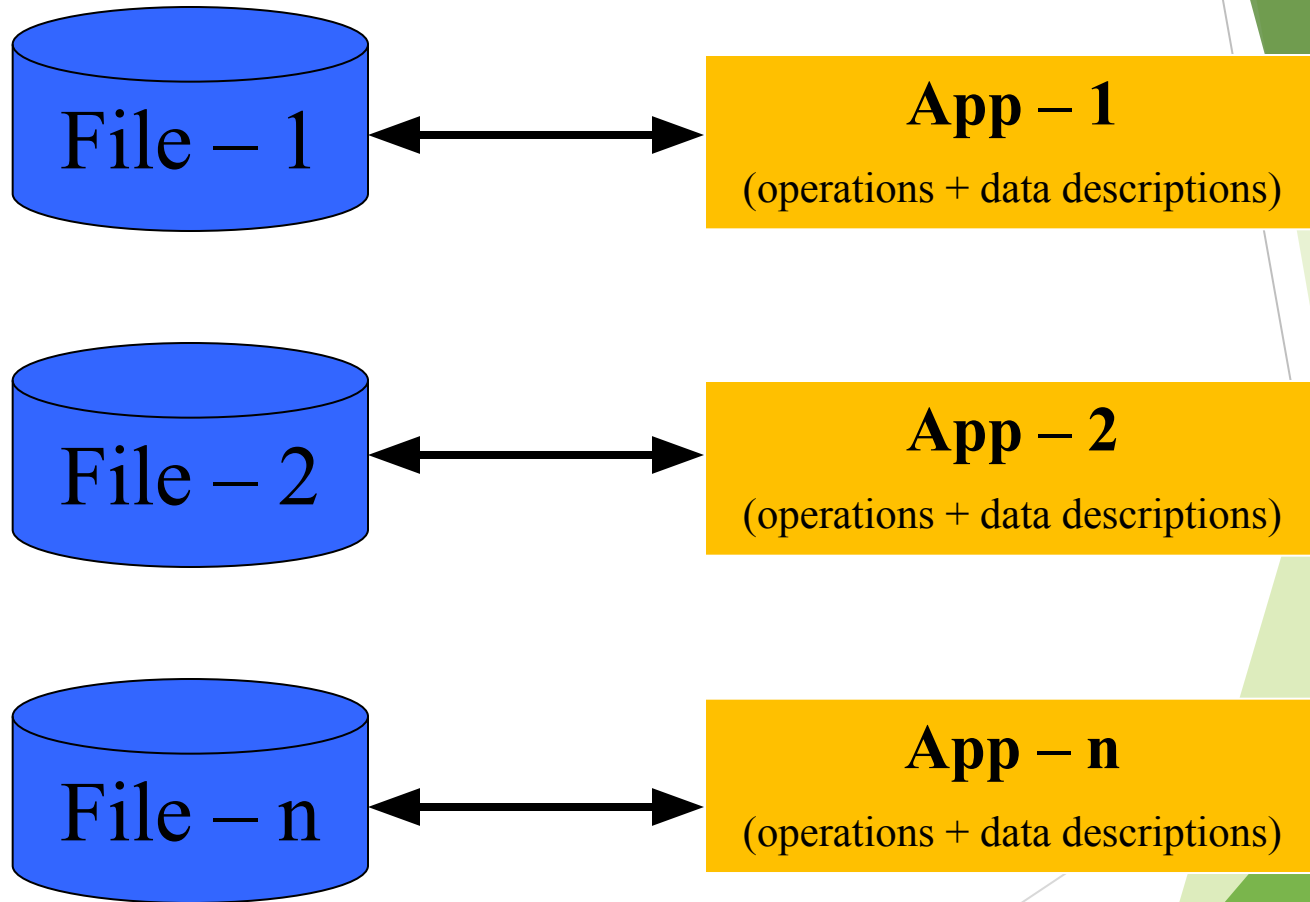
Case Study

- Consider a College System consisting of the following offices:
 - Admission Office
 - Academics Office
 - Exam Office
- Each Office is maintaining its own set of files for its day to day operations.
- Some of the files used in the system are Student's File, Faculty File, Course File, Correspondence File etc.

File Processing System

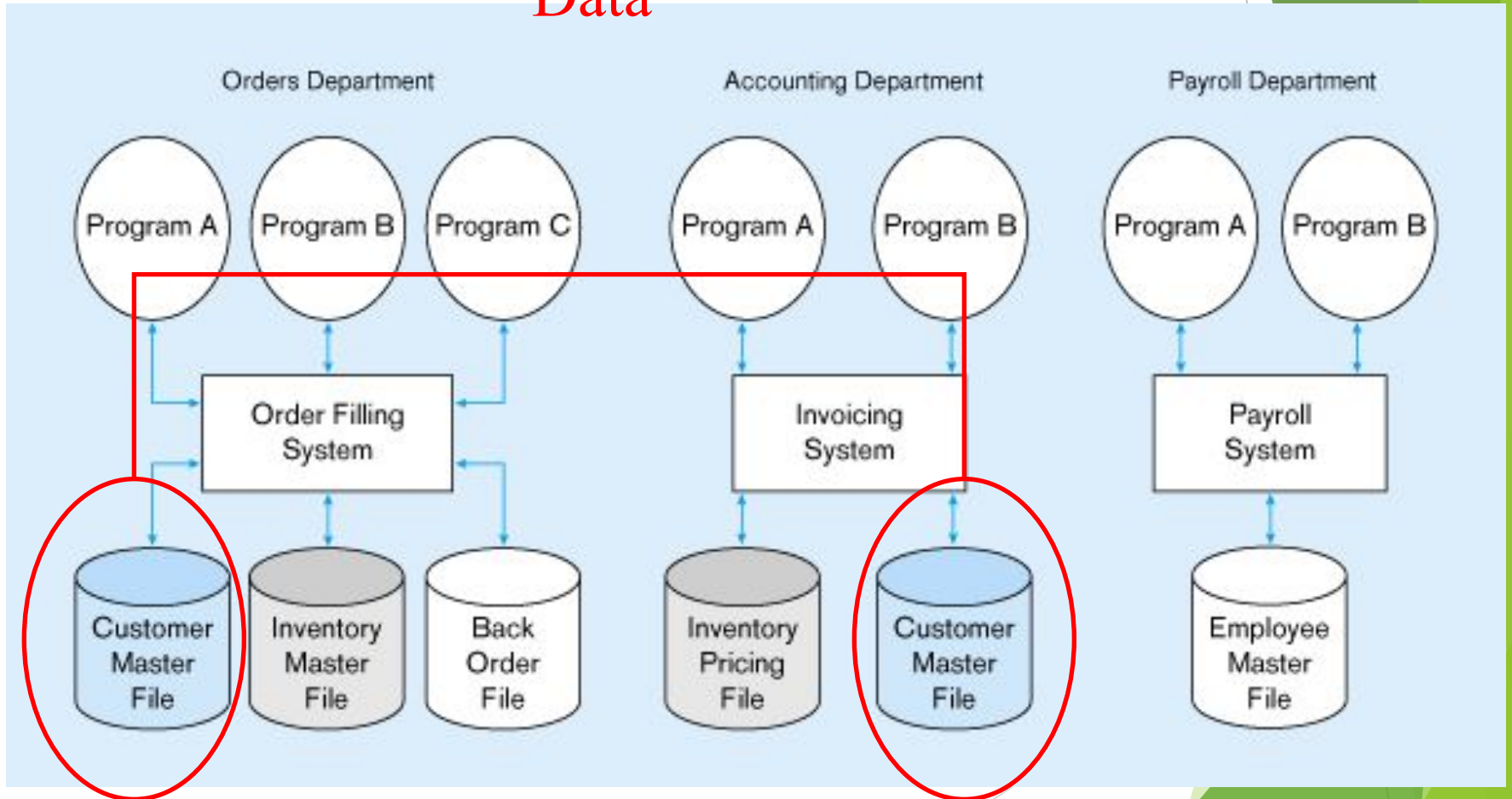
- File-based systems were an early attempt to computerize the manual filing system.
- The manual files were replaced by computer files.
- A person responsible for developing and managing computerized file processing system was called **Data Processing (DP) specialist**.
- In the traditional file processing system, the existing manual system is automated by focusing on the data processing needs of the individual departments instead of treating the organization as whole.
- Each application will have its own set of **Private Files** designed to meet the needs of a particular department.

File Processing System



File Processing System

Duplicate Data



Drawbacks of File System

1. Redundancy of Data
2. Inconsistent Data
3. Poor Enforcement of Standards
4. Data Dependence
5. Limited Data Sharing
6. Inflexible
7. Lack of provision for security
8. Limited recovery from failure

Drawbacks of File System

1. Redundancy of Data

- Redundancy means duplication of data.
- Same information is needed to be kept at different places. It costs time and money.
- Same data may have to be input several times to update all the occurrences.
- Redundancy results in data anomalies (Insert, Update, Delete)

Drawbacks of File System

2. Data Inconsistencies or Data Anomalies

- Redundancy leads to data inconsistency or data anomalies.
- A **data anomaly** occurs if an operation (update, insert, delete) has not yet been performed against all the occurrences. Consequently, same data stored at different places will disagree with each other
- The use of different programming environment and platforms lead to Heterogeneous Environment.

3. Poor Enforcement of Standards

- Organization wide enforcement of standards is poor. This leads to the following types of inconsistencies:
 - a) **Synonym**: Using different names for same data item
Example: Stu-Id, Reg-No.
 - b) **Homonym**: Using same name for different data items.
Example: “course” for a single subject or entire program.

Drawbacks of File System

4. Data Dependence

- The definition of data is embedded in the application programs, rather than being stored separately and independently.
- The applications are constrained to work only with the given file description. Any change in the file structure or data requires changes to all the applications using that file. Such applications are called **Data Dependent Applications**.
- Data dependence leads to excessive program maintenance.
- Often difficult to locate all programs affected by change
- Errors are often introduced when making changes.
- Incompatible file structures (use of different programming languages) makes the situation even worse.

Drawbacks of File System

5. Limited Data Sharing

- As each application has its own private files so little opportunity to share data with others.
- The filing system works well when we have to interact with each file separately. However, it breaks down when we have to **cross-reference** or process the information in the files. This makes it hard to answer queries involving multiple files.

6. **Inflexible:** Fixed queries

7. Lack of provision for security

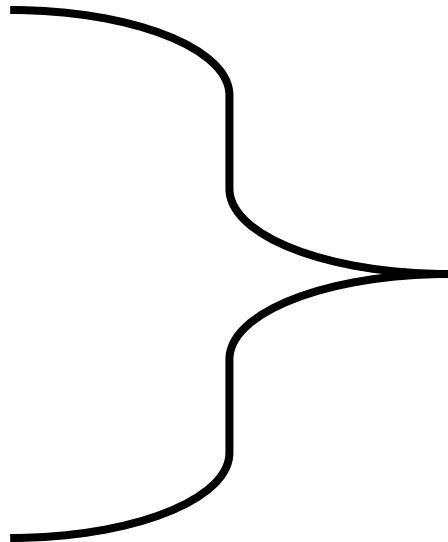
8. Limited recovery from failure

Database

- Database is a collection of information that is organized so that it can be easily accessed, managed and updated.
- Database is a shared collection of logically related data, and a description of this data (metadata), about entities and the relationships among these entities.

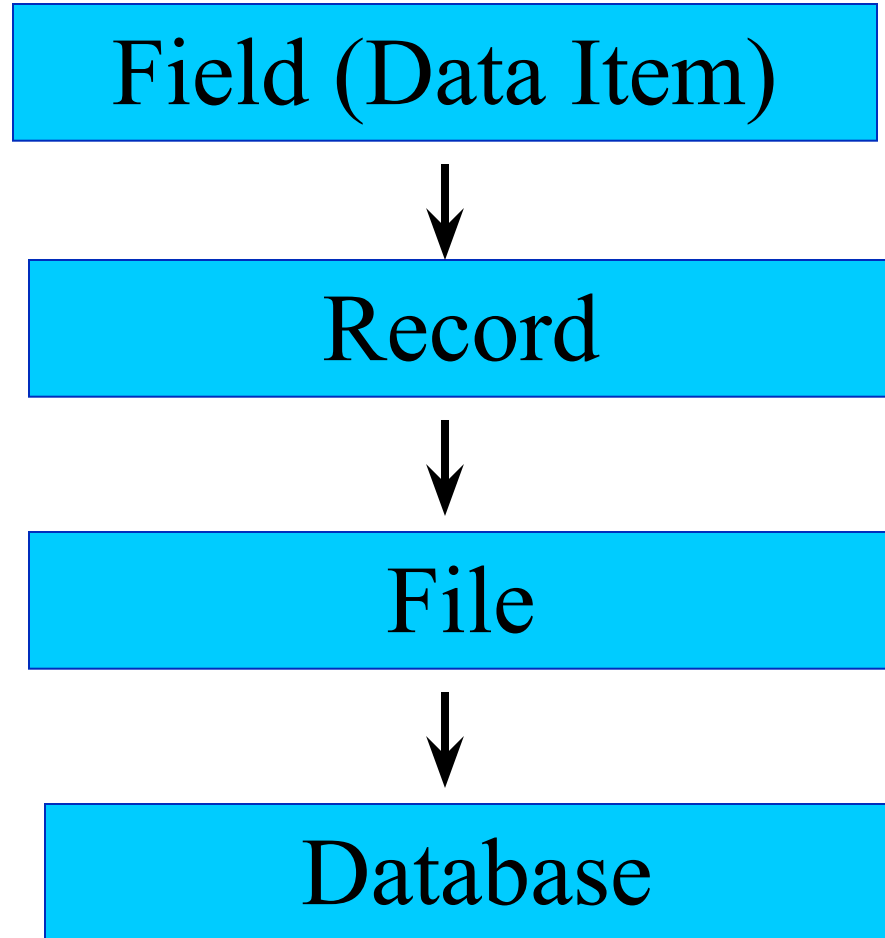
Entities

- ▶ An **entity** is any object in the system that we want to model and store information
- ▶ Each table is about an object, person or thing.
- ▶ Customer
- ▶ Patients
- ▶ Books
- ▶ Products
- ▶ Teachers



**These are known
as entities**

Data Hierarchy



Student Table (Example)

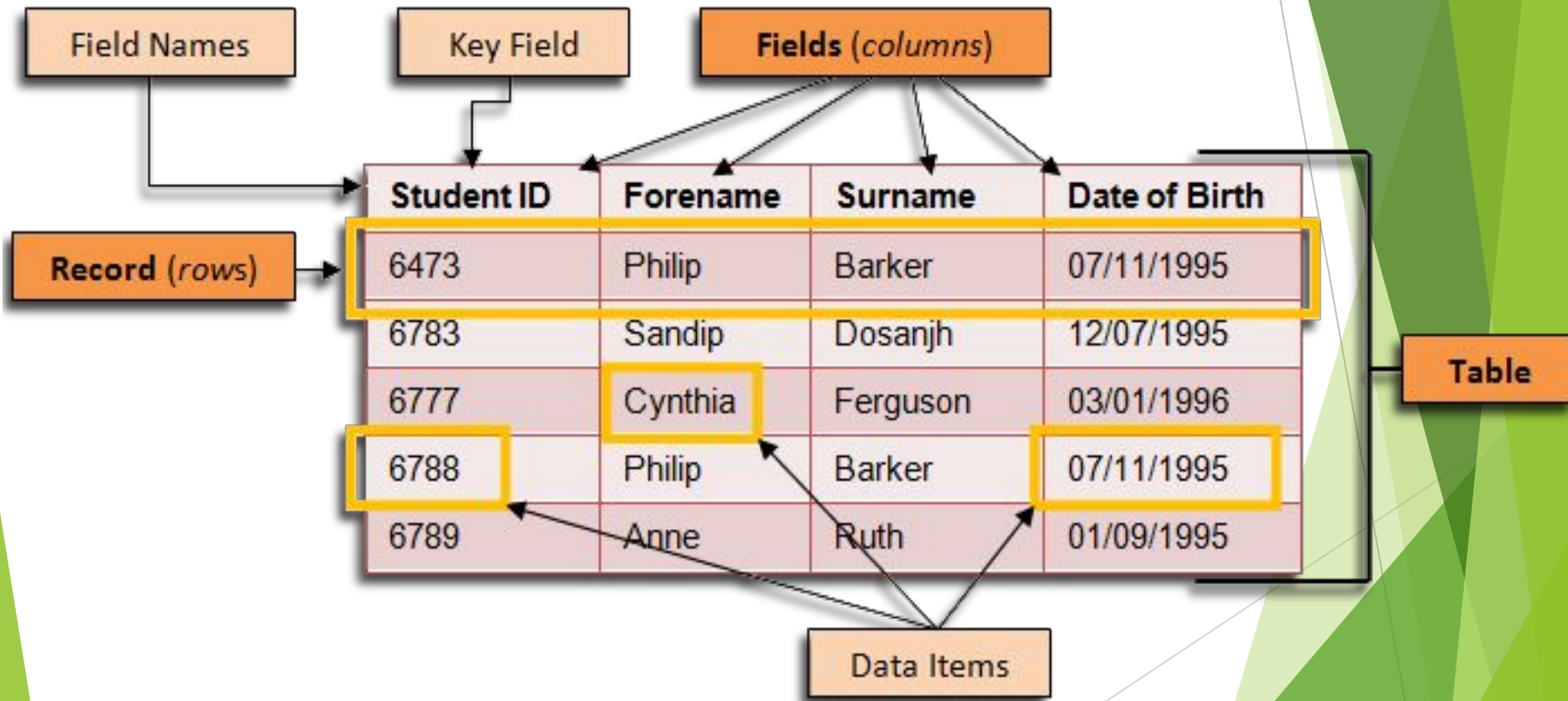


Table Example

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

Metadata

- **Metadata:** Data that describes properties or characteristics of other data OR it is “data about data”.
- The description of data is also known as **system catalog** or **data dictionary**.

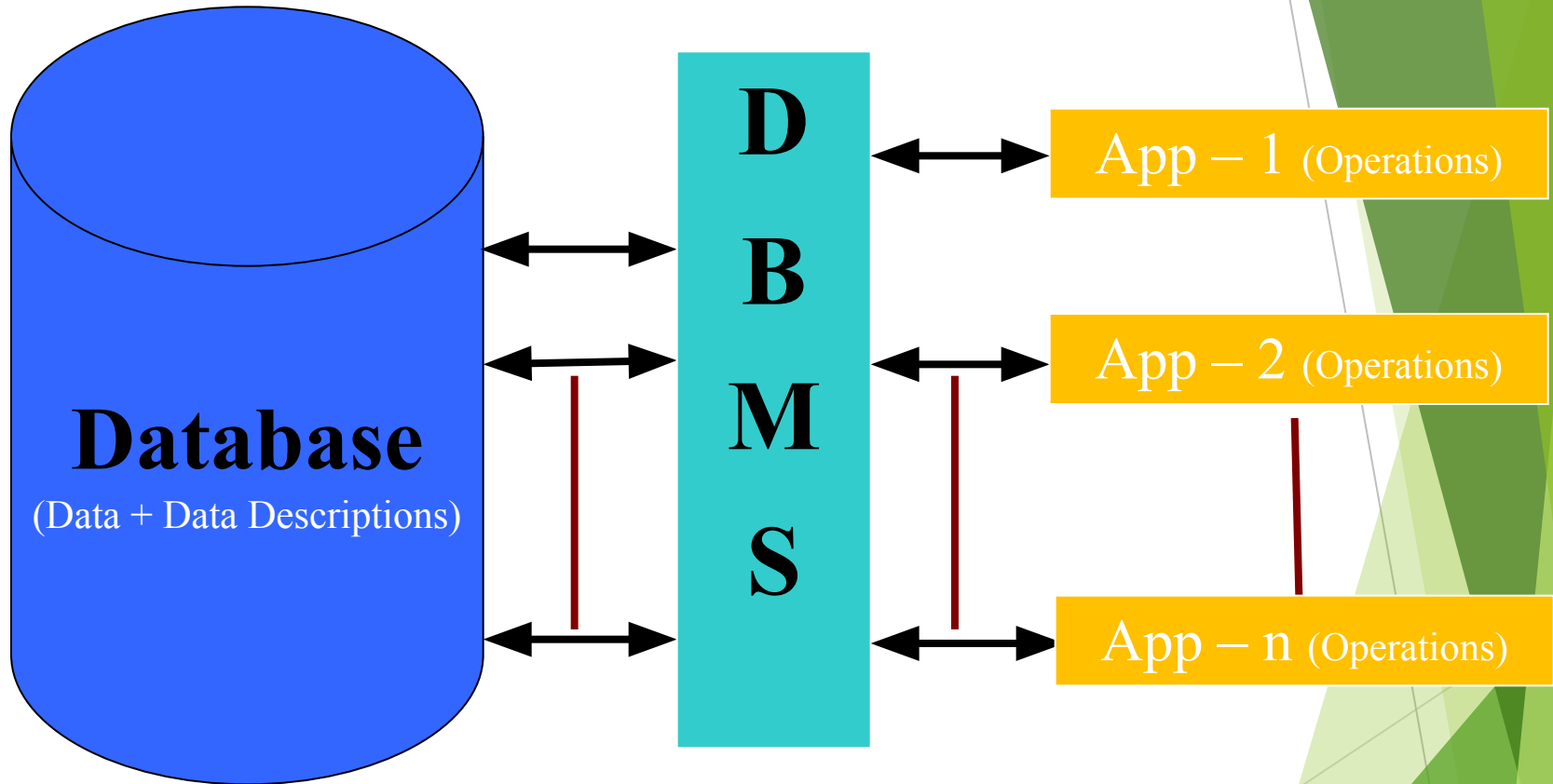
Metadata Example

Data Item		Value			
Name	Type	Length	Min	Max	Description
Course	Alphanumeric	30			Course ID & Name
Section	Integer	1	1	9	Section Number
Semester	Alphanumeric	10	1	8	Semester & Year
Name	Alphanumeric	30			Student name
ID	Integer	9			Student ID
Major	Alphanumeric	4			Student Major
GPA	Decimal	3	0.0	4.0	Student GPA

Examples of Database Applications

- Databases play a critical role in almost all areas where computers are used.
- Databases are everywhere ...
 - Student / Employee's Information System
 - Library System
 - Hotel / Airline Reservation System
 - Billing System
 - Stock / Inventory System
 - Payroll System
 - Geographical Information Systems (GISs)
 - Data Warehouses

Database System



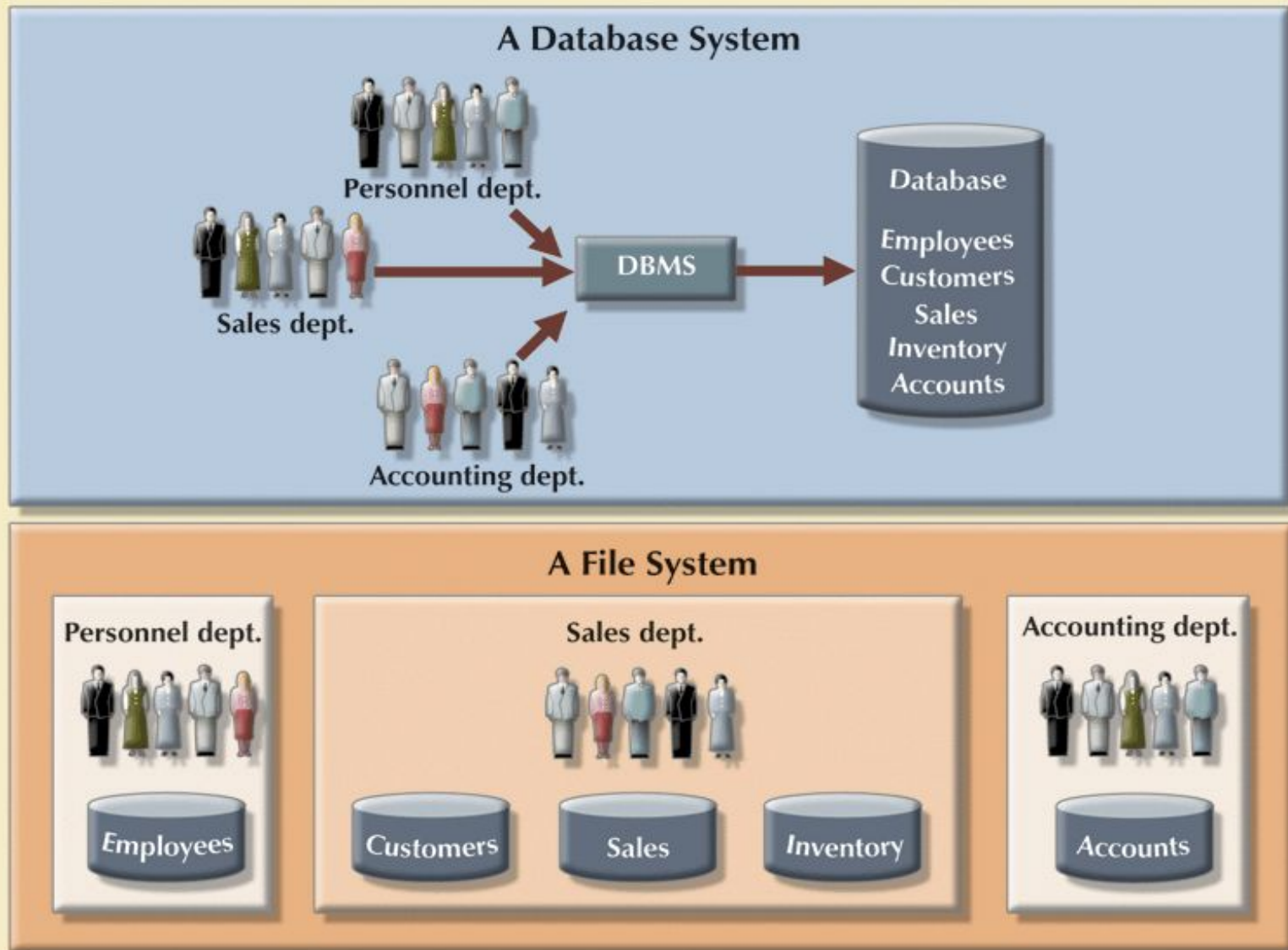
*DBMS manages data resources like
an operating system manages
hardware resources*

Integrated Database Environment(IDE)

- The limitations of file-based approach can be attributed to the following factors:
 - Redundancy of data
 - Data dependence
- In contrast to file processing system, the IDE has a single large repository of data, where data definition is separated from application programs.
- The organization wide requirements are analyzed as a whole and there is no longer concept of MY FILE or Private Files.
- A Database Management System (DBMS) is a software designed to assist in maintaining and utilizing large collections of data.

**FIGURE
1.6**

Contrasting database and file systems



Advantages of Database Approach

1. Reduced or Controlled Data Redundancy
2. Improved Data Consistency
3. Enforcement of Standards
4. Reduced Program Maintenance
5. Data Sharing
6. Data Integrity (Improved Data Quality)
7. Improved Security Restrictions
8. Improved Accessibility & Responsiveness
9. Improved Decision Making
10. Data Independence

Advantages of Database Approach

1. Reduced or Controlled Data Redundancy

- Since all the data items are stored in a single database, therefore Redundancy is minimized.
- Separate data files are integrated into a single, logical structure to reduce redundancy
- 100% elimination of redundancy is not possible in order keep the logical connections among the data items.

2. Improved Data Consistency

- Reduction in redundancy automatically avoids inconsistency.
- Wang and Wand describes data consistency as “a data value can only be expected to be same for the same situation”.
- If a data item appears only once, any change to its value needs to be performed only for once and the database will always be in some consistent / correct state.

Advantages of Database Approach

3. Enforcement of Standards

- This is possible as database is designed to meet the organization wide requirements.
- The standards can be name of data items and their format, data codes, documentation standards, operation standards, security policies etc.

4. Reduced Program Maintenance & Development Cost

- When all the organization's requirements are satisfied by one database instead of many separate files, the maintenance and development cost automatically reduces.

Advantages of Database Approach

5. Data Sharing

- Sharing means that the same data source is used by multiple applications.
- Data is centralized and hence can be shared not only by the existing applications but also new applications can be developed to operate against the same data.
- The applications that can access same data simultaneously are called Online Transaction Processing (OLTP) applications and DBMS uses Concurrency Control Mechanism to ensure that multiple users can access and update data correctly.

Advantages of Database Approach

6. Data Integrity (Improved Data Quality)

- Data integrity refers to the correctness of data.
- Integrity constraints or rules ensure that the data stored in the database is purified and accurate.

7. Improved Security Restrictions

- Database security is the protection of database from unauthorized disclosure, alteration or destruction.
- DBMS provides strong security measures against such threats. Some of them are:
 - Password checks
 - User defined procedures
 - Defining user privileges
 - Audit trial system
 - Data encryption

Advantages of Database Approach

8. Improved Accessibility & Responsiveness.

- In File Processing System, data accessibility is quite difficult as it is procedural based. You should know the detailed procedural steps. HOW TO DO?
- On the other hand, accessing data is lot easier in DBMS with the help of a non procedural language – SQL. You only need to know the simple commands or in other words WHAT TO DO?

9. Improved Decision Making

- Consolidated reports
- Ad-hoc queries
- Multiple views of data

Advantages of Database Approach

10. Data Independence.

- The separation of data descriptions from the applications using the data is called data independence.
- Allows change & evolution of database systems without changing the application programs

Cost & Risk of Database Approach

1. Need for new specialized personnel
2. Installation cost & complexity
3. Conversion from legacy system to modern database system
4. Organizational conflict
5. Higher impact of failure
6. Maintaining currency

Types of Databases

- Databases can be categorized on the basis of the following criteria:

Criteria	Types
1. Number of users	- Single User Database - Multi-user Database
2. Data location	- Centralized Database - Distributed Database
3. Data usage	- Operational - Analytical

Types of Databases

Single user with
desktop computer



Mainframe computer
with thousands of users

- Categories are:
 - Personal computer databases(Single User)
 - Workgroup databases(Multi-user)
 - Department databases (Multi-user)
 - Enterprise database (Multi-user)

Types of Databases

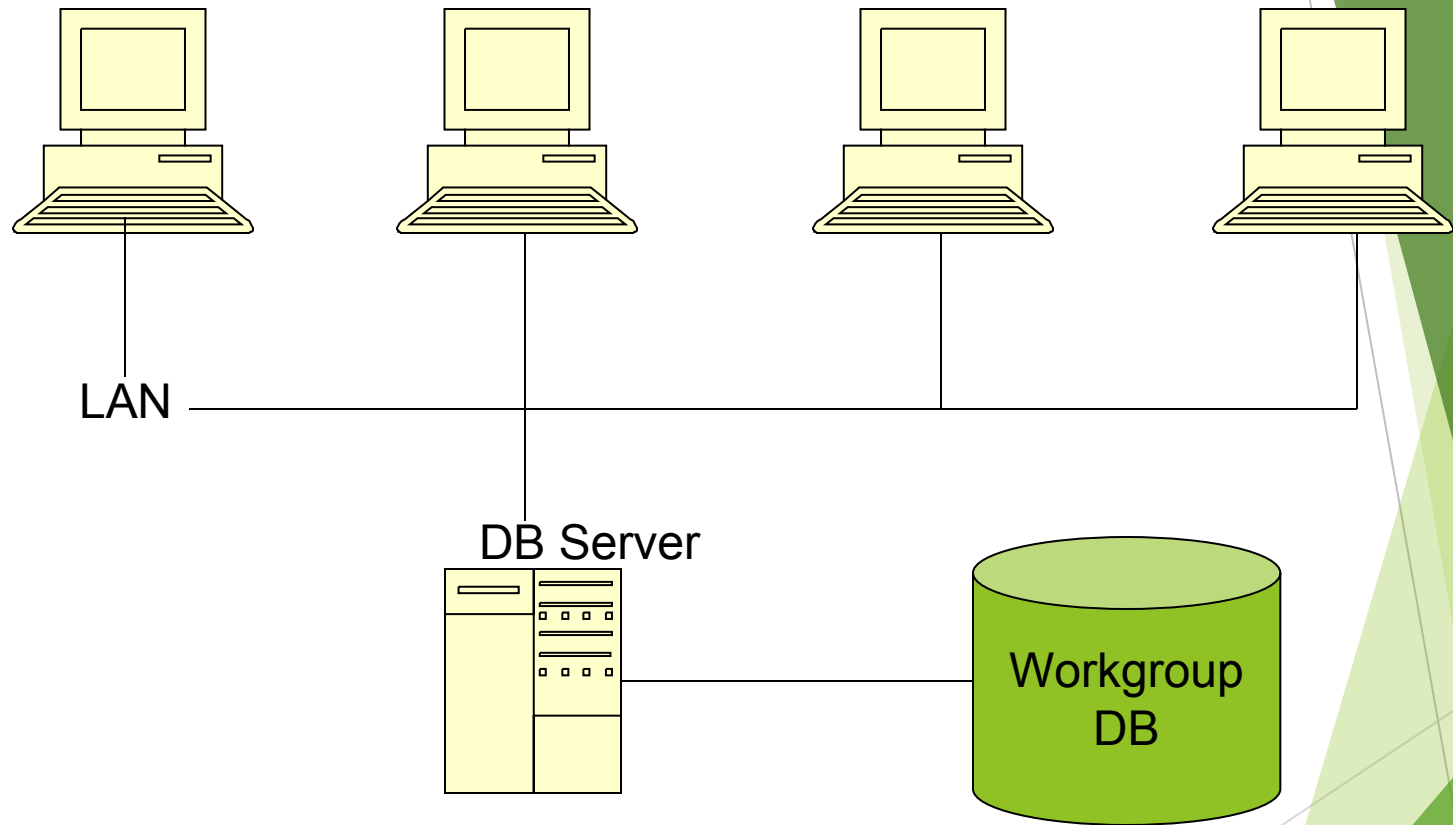
Personal Computer Databases

- Support one user with a standalone PC
- E.g. a student's own database or a sales person's simple database

Workgroup Databases

- Workgroup: relatively small group of people who collaborate on same project/application.
- A workgroup DB supports the collaborative efforts of a workgroup.
- Allows data sharing.
- Its model is shown on the next fig:

Continued...



Workgroup DB on LAN
(Method of Data Sharing)

Types of Databases

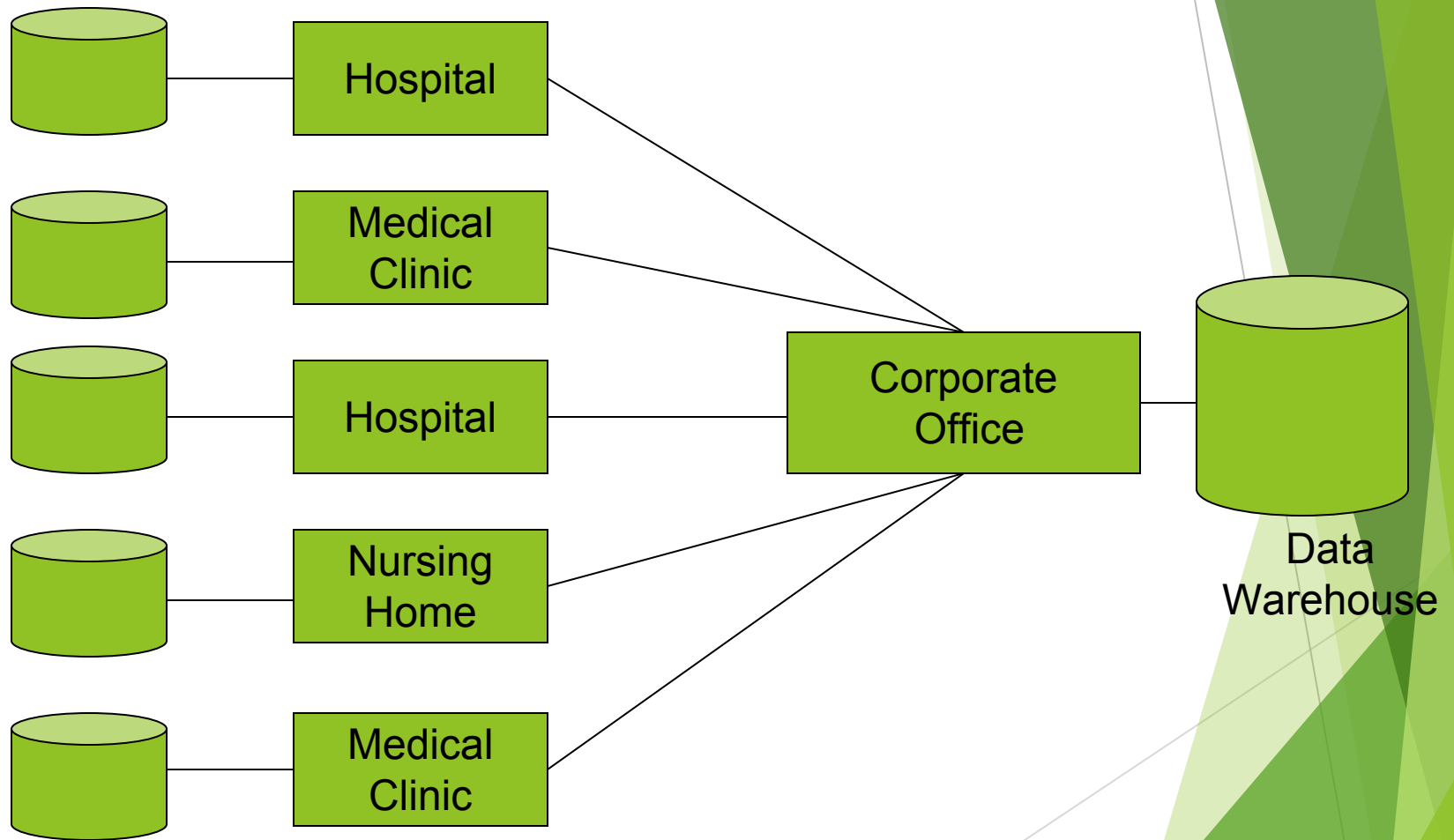
Department Databases

- Department: Functional unit within an organization
- Department DBs support function & activities of a department.
- E.g. personnel DB to track employees, jobs, skills etc.

Enterprise Databases

- DB scope is the entire organization
- To support organization-wide operations & decision making.

Continued...



Types of Databases

Centralized & Distributed Databases

- **Centralized** database system – supports a database located at a single site
- **Distributed** database system – supports a database distributed across several different sites

Types of Databases

- The past few years have produced advances in technology leading to exciting new applications of database systems such as:
 - Multimedia databases
 - Geographic information systems
 - Data warehouses
- Multimedia databases – store pictures, video clips, and sound messages
- Geographic information systems (GIS) – store and analyze maps, weather data, and satellite images
- Data warehouses & on-line analytical processing (OLAP) is used to extract and analyze useful information from very large databases for decision-making.

Reading Material

- ▶ Database Systems – Design, Implementation and Management: Carlos Coronel, Steven Morris (12th Ed.)
- ▶ Chapter 1
 - ▶ Sections 1.1 to 1.6